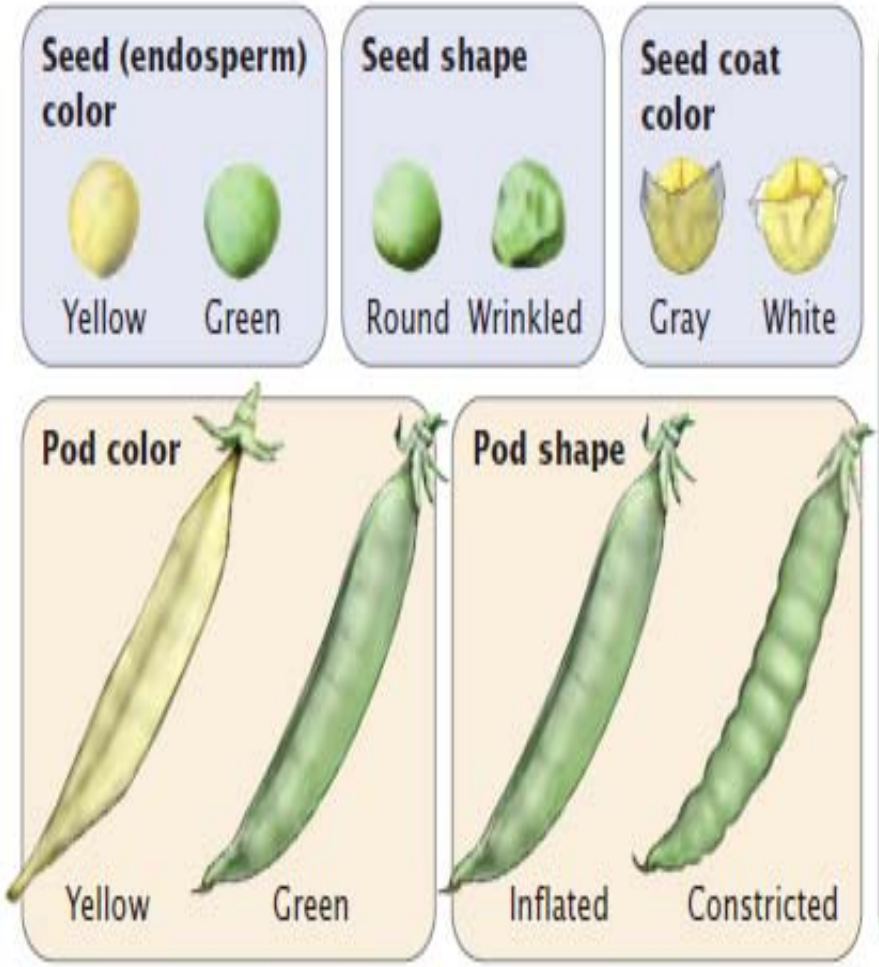
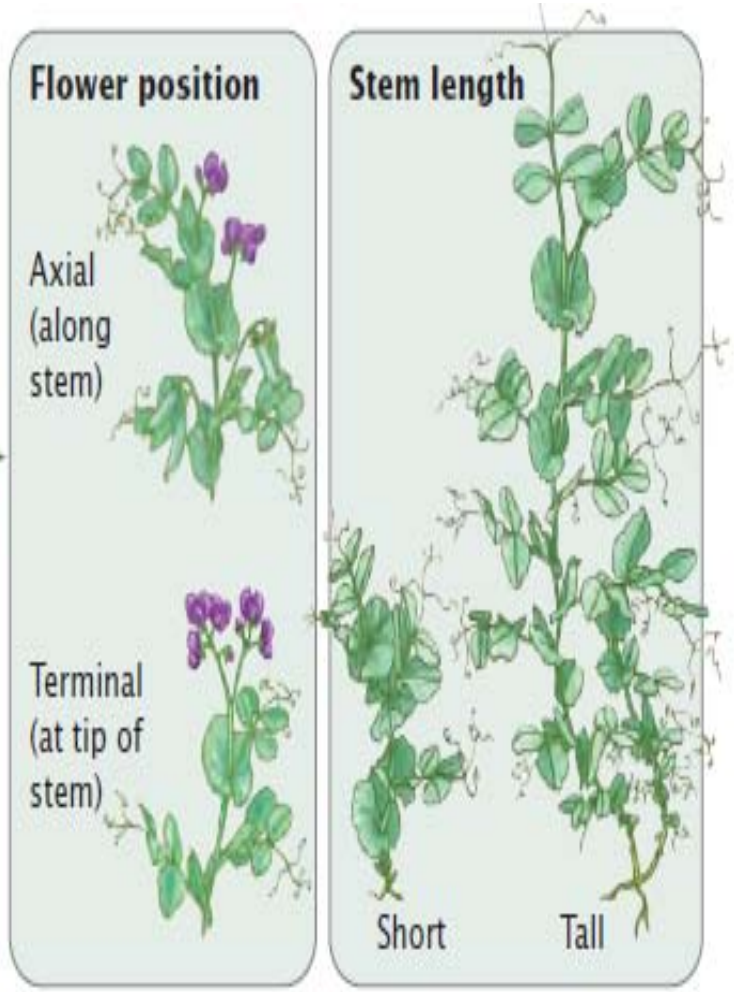
Foundations of Biology - II

Mendelian Genetics

INDRAPRASTHA INSTITUTE *of*
INFORMATION TECHNOLOGY
DELHI

Mendel's paes, *pisum sativum*





Monohybrid cross

- Mendel began by studying **monohybrid crosses—those** between **parents that differed in a single characteristic.**
- In one experiment, Mendel crossed a pure-breeding (homozygous) pea plant for round seeds with one that was pure-breeding for wrinkled seeds.
- This first generation of a cross is the **P (parental) generation.**
- When Mendel examined the F1 generation of this cross, he found that they expressed only one of the phenotypes present in the parental generation: **all the F1 seeds were round**.

F2 generation of monohybrid cross

- In **F2 (filial 2) generation, b**oth of the traits from the P generation appeared.

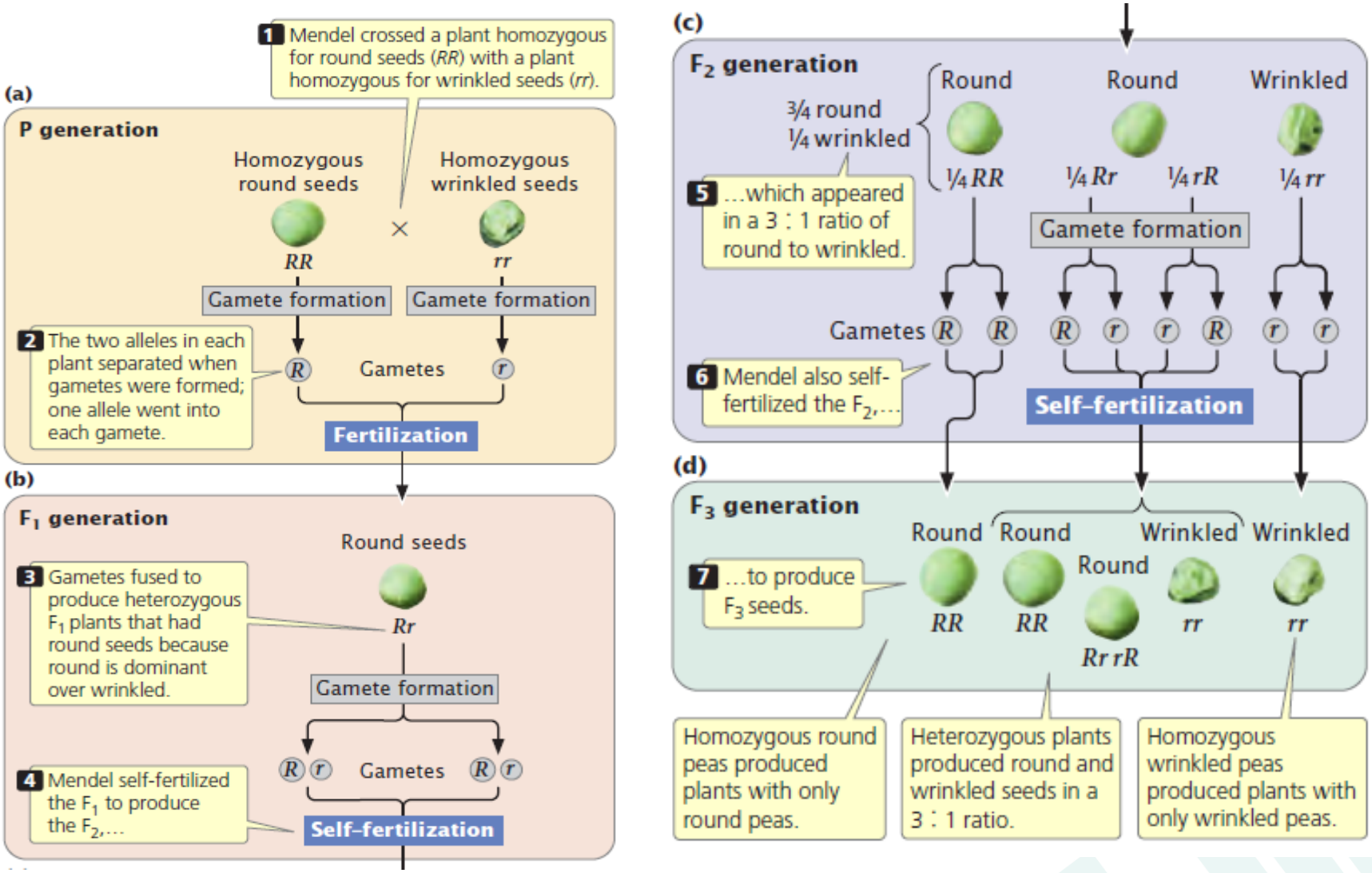
- Mendel counted 5474 round seeds and 1850 wrinkled seeds in the F2.

- The number of the round and wrinkled seeds constituted approximately a 3 to 1 ratio; i.e., in F2 about 3/4 of the seeds were **round** and 1/4 were **wrinkled**.

- Mendel conducted monohybrid crosses for all seven of the characteristics that he studied in pea plants and, in all of the crosses, he obtained the same result:

- All of the F1 resembled only one of the two parents,
- Both parental traits emerged in the F2 in an approximate ratio of 3 : 1.

Mendel's monohybrid cross-flow diagram



What Monohybrid Crosses Reveal?

- **The first conclusion** is that although the F1 plants display the phenotype of only one parent.

  However, F1 generation must inherit genetic factors from both parents because they transmit both phenotypes to the F2 generation.

  Notation: The genetic factors (now called alleles) that Mendel discovered are, by convention, designated with letters; the allele for round seeds is usually represented by *R, and the* allele for wrinkled seeds by *r.*

- **The second conclusion** that Mendel drew from his monohybrid crosses was that the two alleles in each plant separate when gametes are formed, and one allele goes into each gamete **(principle of segregation)**.

Dominant, recessive and gametes

- The **third important** conclusion Mendel derived from his experiment on monohybrid crossest is that of concept of dominance.

  *Those traits that appeared unchanged in the **F1 heterozygous** offspring Mendel called **dominant, and those traits that disappeared in the F1 heterozygous** offspring he called **recessive.***

  When dominant and recessive alleles are present together, the recessive allele is masked, or suppressed.

- Mendel's **fourth conclusion** is that the two alleles of an individual plant separate with equal probability into the gametes.

  When plants of the F1 (with genotype Rr) produced gametes, half of the gametes received the R allele for round seeds and half received the r allele for wrinkled seeds. The gametes then paired randomly to produce the following genotypes in equal proportions among the F2: RR, Rr, rR, rr

Principle of segregation

- The **principle of segregation states that each individual** diploid organism possesses two alleles for any particular characteristic.

  These two alleles segregate (separate) when gametes are formed, and one allele goes into each gamete.

- **The concept of dominance** states that, when the two alleles of a genotype are different, only the trait encoded by one of them—the "dominant" allele—is observed.

Meiosis and Mendel's principle

Comparison of the principles of segregation and independent assortment

Principle	Observation	Stage of Meiosis
Segregation (Mendel's first law)	1. Each individual organism possesses two alleles encoding a trait.	Before meiosis
	2. Alleles separate when gamets are formed.	Anaphase I
	3. Alleles separate in equal proportions.	Anaphase I
Independent assortment (Mendel's second law)	Alleles at different loci separate independently.	Anaphase I

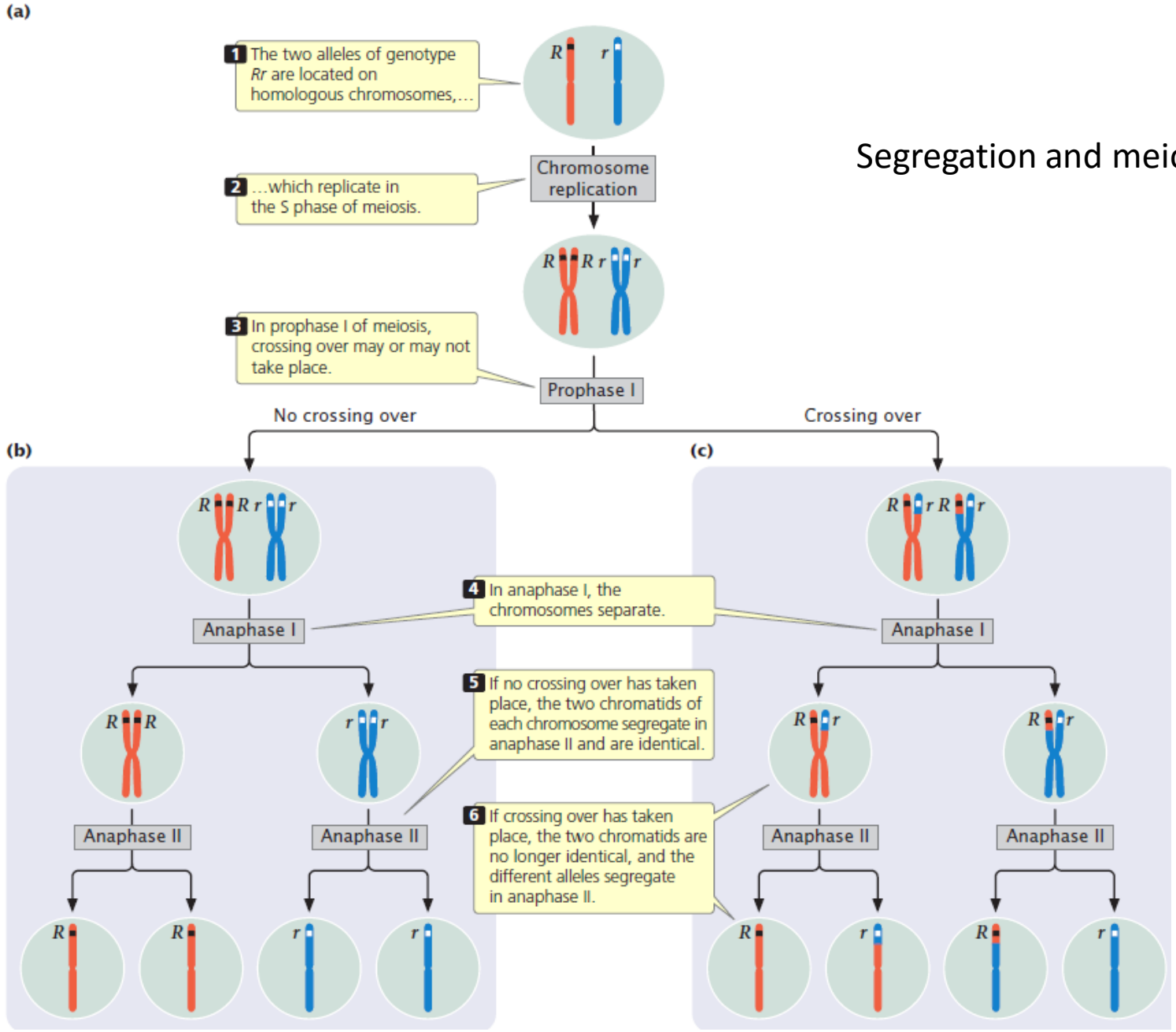
Concept check.

- How did Mendel know that each of his pea plants carried two alleles encoding a characteristic?

- Draw the cell and show that Segregation results from the separation of homologous chromosomes in meiosis. Draw all the steps involved in it.

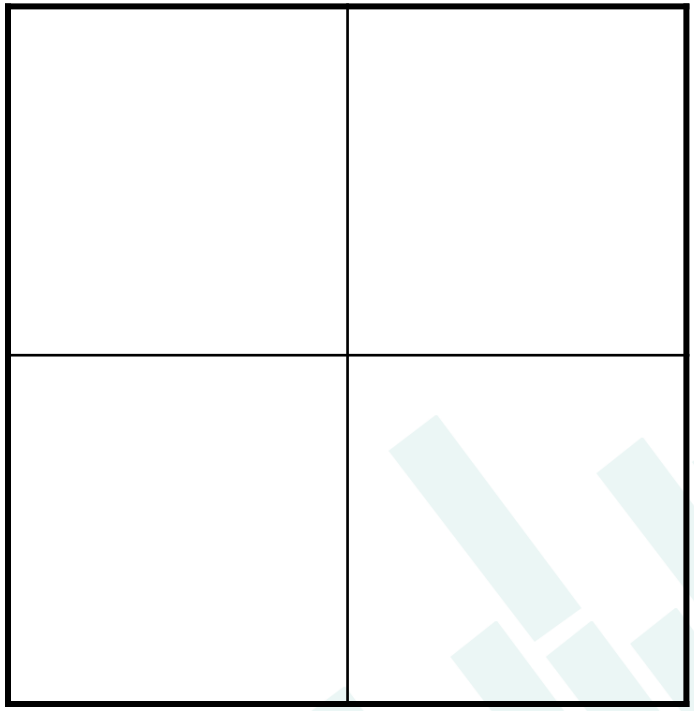
  (see next slide).

Segregation and meiosis



Predicting the outcomes of Genetic Crosses- Punnett squares

- A useful tool to do genetic crosses.
- For a **monohybrid cross**, you need a square divided by four….
- Looks like a window pane…
- Punnett square is used to predict both the genotypes and phenotypes of the offspring.



Steps to use Punnett square

STEPS:

1. Determine the genotypes of the parent organisms
2. Write down your "cross" (mating)
3. Draw a p-square

Example:

Parent genotypes:

TT and *t t*

Cross

T T × ***t t***

Continue

4. **“Split" the letters of the genotype for each parent & put them "outside" the p-square**
5. Determine the possible genotypes of the offspring by filling in the p-square
6. Summarize results (genotypes & phenotypes of offspring)

T T × *t t*

(both pure Cell-lines)

	T	**T**
t	**T *t***	**T *t***
t	**T *t***	**T *t***

Genotypes:
100% T t

Phenotypes:
100% Tall plants

Monohybrid cross: F_2 generation

- If you let the **F1 generation self-fertilize**, the next monohybrid cross would be:

T *t* × **T *t***

(tall) (tall)

	T	***t***
T	**T T**	**T *t***
t	**T *t***	***t t***

Genotypes:

1 TT= Tall

2 T*t* = Tall

1 *tt* = dwarf

Genotypic ratio= 1:2:1

Phenotype:

3 Tall

1 dwarf

Phenotypic ratio= 3:1

Backcross and testcross

- The cross between an F1 genotype and **either of the parental genotypes**, is called a backcross.
- In the **test cross**, always the parental genotype is **recessive**.

Dihybrid cross

- Mendel crossed varieties of peas that differed in *two characteristics—a* **dihybrid cross**.

- For example, Mendel had one homozygous variety of pea with seeds that were round and yellow; another homozygous variety with seeds that were wrinkled and green.

- When he crossed the two varieties, the seeds of all the F1 progeny were round and yellow.

- Self fertilization of F1 results in progeny in the F2 and the traits appeared approximately in a 9 : 3 : 3 : 1 ratio.

- **An important qualification of the principle of independent assortment is that it applies to characters encoded by loci located on different chromosomes because, like the principle of segregation, it is based wholly on the behavior of chromosomes in meiosis.**

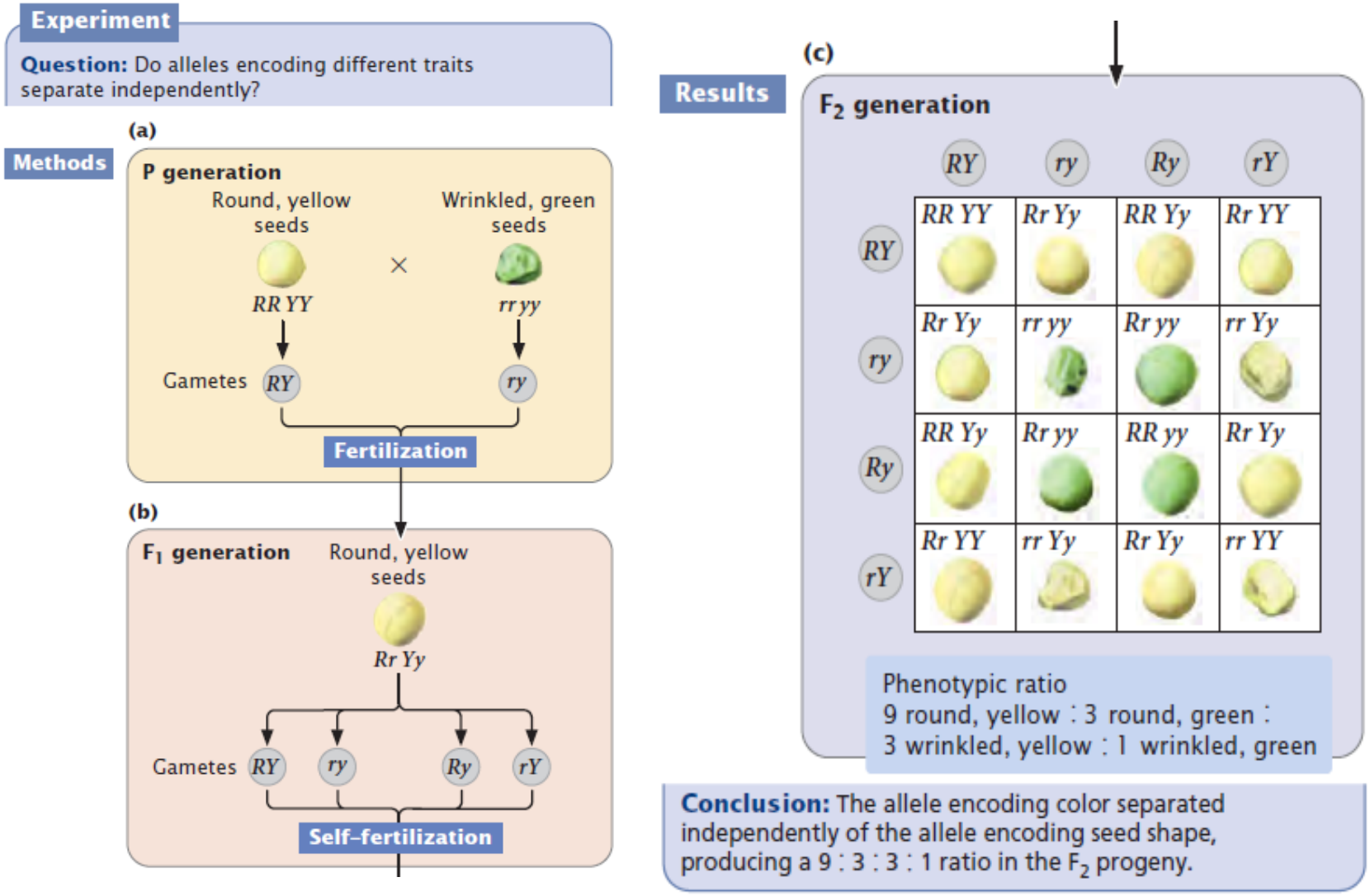
The Principle of Independent Assortment

- The **principle of independent assortment (Mendel's second law)** states that **alleles at different loci** separate independently of one another.

- Principle of independent assortment is actually an extension of principle of segregation.

- **Summary:** The **principle of segregation** states that the two alleles of a locus separate when gametes are formed; the **principle of independent assortment** states that, when these two alleles separate, their separation is independent of the separation of alleles at *other loci.*

Explanation of principle of independent assortment



Concept check

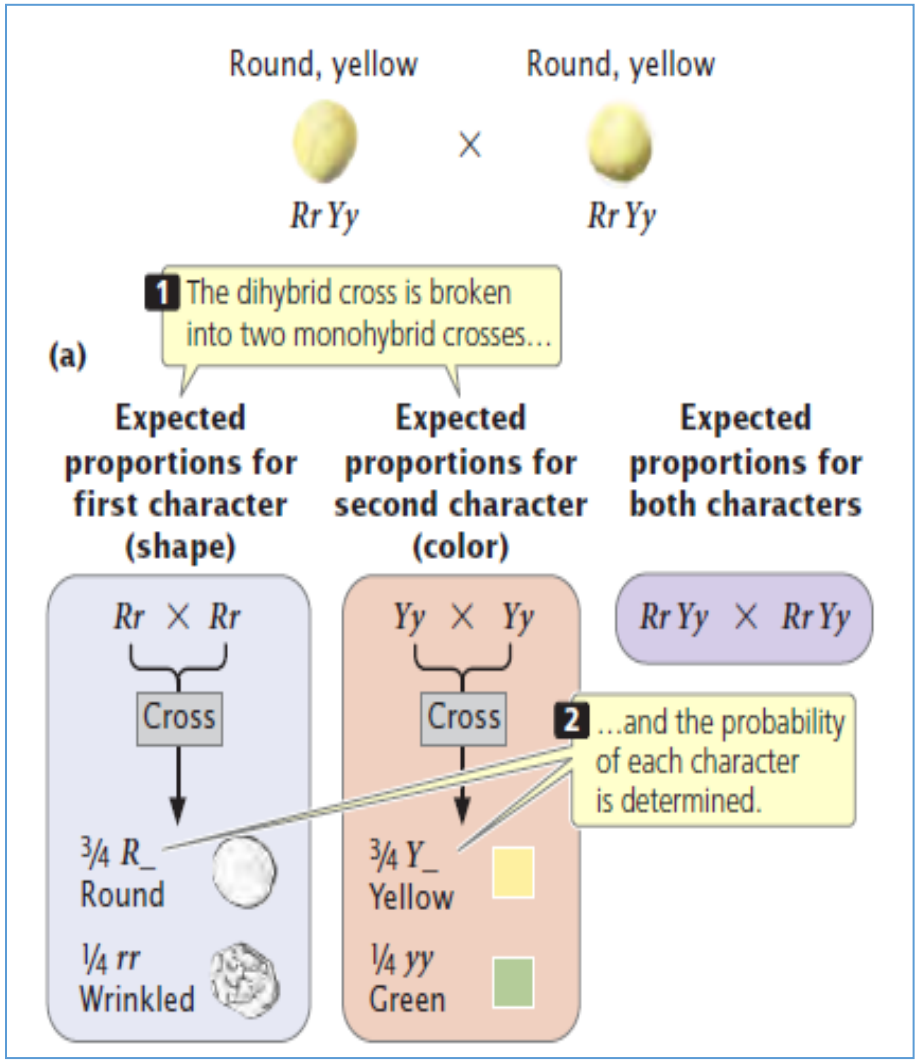
❖How are the principles of segregation and independent assortment related and how are they different?

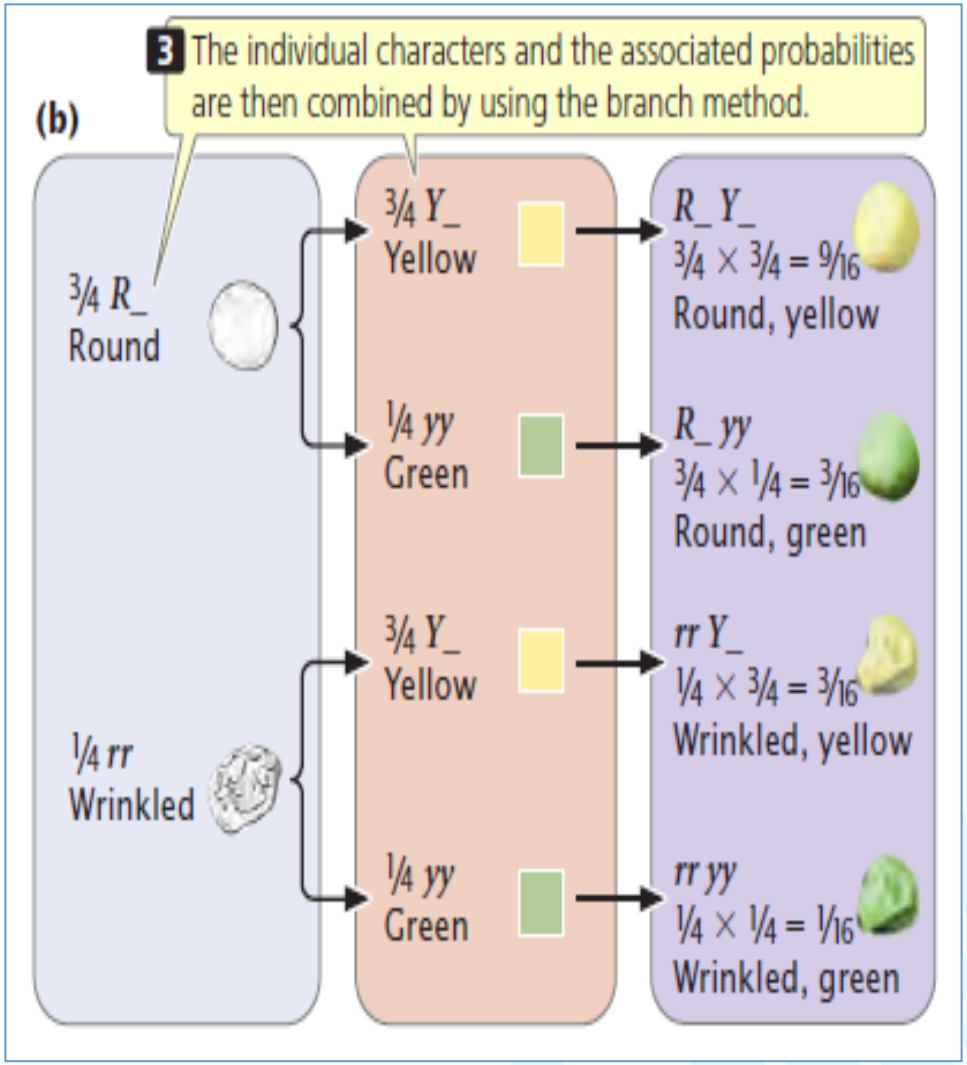
- Answer

❖The principle of segregation and the principle of independent assortment both refer to the separation of alleles in anaphase I of meiosis.

❖The principle of segregation says that these alleles separate, and the principle of independent assortment says that they separate independently of alleles at other loci.

(a) Branch diagram for dihybrid cross





(b) Alternate listing method

- This method helps to determine how many gametes are possible in the heterozygous loci.
- Suppose if ‘n’ is the number of loci that are heterozygous, the number of gametes will be equal to $2^{n.}$
- Take for example the double heterozygous Aa Bb.
- There are two heterozygous loci, and there will be $2^2 = 4$ different types of gametes.
- Of the four gametes, 2 will have allele A and 2 will have allele a. List these like this:
- A A a a

Continued

- To introduce Bb, the following steps are taken:
- Of the gametes with allele A, half will be ‘B’ and half will be ‘b’.
- Likewise, half of those with ‘a’ allele, will have ‘B’ and half will have ‘b’.
- So now write like this

A A a a
B b B b

- Solve: In each of the following cases, use alternate list method to determine the kinds and proportions of gametes that the individual will produce.

(a) Mm Nn (b) AA bb (c) RR Tt

Summary: Six systematic steps to identify geneotypes and phenotypes

- **Step A:** Make a diagram: Use standard designations for the generations and lay out the problem in simple diagram.

- **Step B:** Choose Symbols: Make a key to your symbols.

- **Step C:** Discover parental information; i.e., determine the genotypes of the parents in each cross.

To do this use evidence: Are the parent true breeding lines? Can their genotypes be deduced from their phenotypes? Do the phenotype of their offspring provide any information?

Continued

- **Step D:** Determine the parent Gamete types. Indicate the possible type of gametes formed by each of the parents.

  If it's a monohybrid cross, use the law of segregation.

  If it's a dihybrid cross, use the law of independent assortment.

- **Step E:** Set up the Punnett Square.

  Set up the Punnet square to show all the zygotes.

- **Step F:** Fill in the Punnett Square.

  Fill it to determine the genotypic and phenotypic ratios.

Given the data about offspring, make conclusions about the parents

If the offspring ratio is:	Then we know this about the parents	The cross is called
3:1	Both parents are heterozygous at one locus. One allele is dominant (Aa X Aa)	Monohybrid cross
1:1	One parent is homozygous recessive, One parent is heterozygous (Aa X aa)	Test cross
9:3:3:1	Each parent is heterozygous at two loci. One allele is dominant at each locus. (Aa Bb X Aa Bb)	Dihybrid cross
1:1:1:1	One parent is double heterozygote. One parent is homozygous recessive for both loci (OR)	Two-point test cross
1:1:1:1	One parent is heterozygous at the first locus and homozygous at the second. One parent is heterozygous at the second locus and recessive at the first (Bb tt X bb Tt)	